

Energy-to-Token Production Revisited: How Numerical Precision, Batch Size, and Context Length Jointly Determine LLM Inference Energy Efficiency

An Empirical Study on Consumer/Mid-Tier GPU Hardware

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Abstract

On a single 1.1-billion-parameter model (TinyLlama-1.1B-Chat) and a single, widely available, non-datacenter-class GPU (NVIDIA T4, as provisioned by Google Colab), this study tests, as one controlled data point in an active and still-unsettled debate, the common assumption that reducing numerical precision through quantization reliably reduces the energy cost of large language model (LLM) inference. A full factorial design of three precision levels (FP16, INT8, INT4), four batch sizes (1, 4, 8, 16), and three context lengths (128, 768, and 1792 tokens) was evaluated, with eight repeated measurements collected for each configuration. Thirty-three of the thirty-six planned configurations completed successfully; the remaining three failed with out-of-memory errors under a consistent, precision-independent pattern. Contrary to the commonly assumed default, and consistent with at least one other single-GPU controlled study, unquantized FP16 achieved the lowest energy per output token in nearly every tested configuration, while INT8 was consistently the least efficient precision level. The benefit of batching was found to be non-uniform and to interact with context length in ways not predictable from either factor in isolation: a substantial batching benefit at short context weakened at medium context length and reversed into a net energy penalty for FP16 at the longest context length tested. These findings are reported using an end-to-end energy-per-token metric, adopted after an earlier attempt to isolate decode-phase energy through subtraction was found to be statistically unreliable at this workload scale. The results are interpreted in relation to prior and concurrent work on quantization, batching, and inference energy measurement — including very recent large-scale, multi-GPU studies that reach a more favorable view of quantization in memory-constrained deployments — and are presented explicitly as setup-specific empirical evidence from a single model and a single GPU generation, not as a general physical claim superseding that broader literature.

Keywords: LLM inference; energy efficiency; quantization; batching; GPU power measurement; Green AI

1. Introduction

1.1 Background, Context, and the Research Problem

The rapid adoption of large language models (LLMs) has shifted the center of gravity of machine learning’s energy cost from training toward inference: a model is trained once but queried potentially billions of times over its deployment lifetime, and Argerich and Patiño-Martínez [1] estimate that for widely used models the cumulative inference energy can exceed training energy within the first month of deployment. This shift has coincided with growing regulatory and public attention to AI’s environmental footprint, reflected in the systematic review by Ji and Jiang [2], which documents rising electricity demand attributable to large language models and calls explicitly for more granular, reproducible measurement rather than vendor-reported or extrapolated figures. In response, a set of engineering practices has become the default toolkit for cheaper, faster LLM inference: reducing numerical precision through post-training quantization, increasing batch size, and, more recently, restructuring key-value (KV) cache memory management [7], [13]. These practices are widely believed to reduce energy consumption as a natural consequence of reduced memory footprint or increased throughput, yet the specific claim that lower numerical precision straightforwardly translates into lower energy consumption per output token has rarely been tested directly, end-to-end, on hardware outside the datacenter-class accelerators for which most quantization kernels are designed and benchmarked.

The research problem this study addresses follows directly: it is not established, for an LLM deployed on a widely accessible, mid-tier GPU rather than a flagship datacenter accelerator, whether reducing numerical precision reliably reduces energy consumption per output token, or whether this reduction is contingent on other operating conditions such as batch size and input context length. This matters practically because most academic researchers, students, and small organizations working with LLMs rely on GPUs such as the NVIDIA T4 rather than an A100 or H100; if quantization’s energy behavior on this class of hardware differs materially from what is reported on flagship accelerators, guidance drawn from the latter may actively mislead practitioners deploying on the former.

1.2 Research Gap

The related work reviewed in Section 2 shows that, while individual optimizations — quantization, batching, and memory-efficient attention — have each been studied extensively in isolation, and while broad reviews of AI energy consumption call for more measurement [2], to the author’s knowledge no controlled, single-GPU factorial study jointly varies precision, batch size, and context length while measuring end-to-end energy per output token directly through hardware power sampling rather than through proxy metrics such as FLOP counts or latency alone. Argerich and Patiño-Martínez [1] come closest, directly measuring energy on real hardware and reporting that quantization does not uniformly reduce energy consumption; however, their study does not vary context length as an experimental factor, leaving open the question of whether the batching and quantization effects they observe are stable as input sequences grow substantially longer, which is precisely the condition under which key-value cache pressure becomes most

severe according to systems-level work on LLM serving [7], [13]. Section 2.6 discusses several large-scale, multi-GPU measurement studies published concurrently with this work that bear on this gap without closing it, and qualifies the claim above accordingly; with that qualification in place, the controlled single-GPU factorial design remains the specific gap the present study addresses.

1.3 Research Objectives and Questions

The primary objective of this study is to determine, through controlled and repeated measurement on a single consumer/mid-tier GPU, how numerical precision, batch size, and context length jointly affect the energy efficiency, latency, throughput, memory footprint, and output quality of LLM inference. This objective is expressed as the following research question: how do numerical precision, batch size, and context length jointly affect the energy efficiency, measured in joules per output token, of LLM inference, and does quantization reliably reduce energy consumption, or is its effect contingent on these other factors? Four hypotheses follow directly from this question and from the literature reviewed in Section 2. The first hypothesis is that quantization does not necessarily reduce energy consumption per output token, and that its effect depends on precision level, batch size, and context length, potentially being dominated by dequantization overhead on hardware not well optimized for low-precision kernels, consistent with the mechanism reported by Argerich and Patiño-Martínez [1]. The second hypothesis is that the energy benefit of batching is non-monotonic in batch size, with returns diminishing as GPU utilization approaches saturation. The third hypothesis is that context length interacts with both precision and batch size such that the relative ranking of configurations by energy efficiency is not stable across context lengths, a possibility raised but not directly tested by the batch-size analysis in Argerich and Patiño-Martínez [1], whose experiments used prompts of fewer than ten tokens. The fourth hypothesis is that the dominant resource constraint, whether a configuration is compute-bound or power-bound, shifts across the experimental matrix rather than remaining fixed for a given model and GPU.

1.4 Significance of the Study

This study is significant for two audiences. For practitioners and researchers who deploy or experiment with LLMs on non-datacenter-class hardware, the findings provide direct, measured evidence about when quantization helps and when it does not, rather than guidance extrapolated from flagship-accelerator benchmarks. For the broader Green AI research agenda articulated by Schwartz et al. [6] and grounded empirically by Strubell et al. [5], this study contributes a controlled dataset that jointly varies three factors rather than one, making explicit the interaction effects that a single-factor study cannot reveal, and it documents a measurement pitfall — the unreliability of differential prefill/decode energy estimation at short-generation scales — that is likely to recur in other energy-measurement studies using similarly short-duration workloads.

1.5 Overview of the Paper

The remainder of this paper is organized as follows. Section 2 reviews prior work on inference energy measurement, quantization, batching, and memory-efficient serving, and

identifies the specific gap this study addresses. Section 3 describes the experimental methodology, including the measurement pipeline illustrated in Figure 1, the energy-integration procedure, the methodological correction described in Section 3.6, and the statistical treatment applied to the results. Section 4 presents the results of the thirty-three completed configurations. Section 5 interprets these results in depth and situates them relative to the literature reviewed in Section 2. Section 6 concludes and states the study’s contributions explicitly. Section 7 offers recommendations and directions for future work, and Section 8 discusses the limitations that bound the generalizability of the findings.

2. Literature Review

2.1 Energy and Environmental Cost of Machine Learning

Concern over the energy and carbon cost of large-scale machine learning was brought into mainstream research attention by Strubell et al. [5], who quantified the financial and environmental cost of training several large NLP models and argued that accuracy improvements were increasingly purchased at a disproportionate energy cost; a key limitation of that study, which the authors themselves note, is that it focuses on training rather than inference, and that its energy estimates rely on published hardware power ratings rather than direct, continuous power measurement during execution. Schwartz et al. [6] extended this concern into a broader methodological argument, proposing “Green AI” as a research value in which computational efficiency is reported and optimized alongside accuracy rather than treated as an afterthought; their contribution is primarily conceptual and does not itself provide a measurement methodology, which is the gap the present study’s energy-measurement pipeline is intended to help fill for the specific case of LLM inference. Patterson et al. [11], writing from an industrial perspective, showed that operational best practices such as efficient model architecture selection and data-center siting could reduce the carbon footprint of large-scale training substantially; their analysis, however, is again focused on training rather than inference, and depends on Google-internal operational data that is not independently reproducible by external researchers, which is a limitation the checkpointed, open, and reproducible measurement pipeline used in this study is designed to avoid.

2.2 Quantization Methods and Their Assumed Energy Benefit

Jacob et al. [4] introduced the integer-arithmetic-only quantization scheme that underlies much of the eight-bit quantization tooling used in practice today, demonstrating that carefully calibrated integer inference could match floating-point accuracy while executing on integer-only hardware; their evaluation, however, is conducted on image-classification convolutional networks rather than autoregressive transformer language models, and does not report end-to-end wall-clock energy measurements, focusing instead on latency and accuracy. Han et al. [12] proposed a three-stage compression pipeline combining pruning, quantization, and Huffman coding, and reported large storage reductions without accuracy loss; this line of work, like Jacob et al. [4], is evaluated on convolutional vision architectures and predates the autoregressive decoding workload pattern that dominates LLM inference,

in which the KV cache rather than the model weights alone plays a central role in resource consumption. In the LLM setting specifically, Dettmers et al. [8] introduced the eight-bit matrix-multiplication procedure adopted by the bitsandbytes library used in the present study, showing that models with up to 175 billion parameters could be quantized to INT8 with negligible accuracy degradation; their evaluation, however, centers on memory footprint and accuracy rather than on measured energy consumption, and does not report whether the additional dequantization computation required at each layer might offset the memory-related savings in terms of energy, which is a question the present study answers directly for a smaller-scale model on non-datacenter hardware. Frantar et al. [9] proposed GPTQ, an approximate second-order one-shot weight-quantization method that pushes precision down to three or four bits with limited accuracy degradation, primarily targeting the practical problem of fitting very large models onto a single GPU; energy consumption is not a reported metric in that work. Gholami et al. [17] provide a broad survey of quantization methods for efficient neural network inference and explicitly note, as a limitation of the field rather than of their own survey, that most quantization research reports accuracy and memory or latency figures without reporting directly measured energy consumption, which the present study addresses by measuring energy directly via hardware power sampling rather than inferring it from latency or FLOP counts. Taken together, this body of work establishes that quantization methods reliably reduce memory footprint and, in most reported settings, either preserve accuracy or degrade it only slightly; it does not, however, establish that quantization reliably reduces energy consumption per output token, a distinction the present study’s direct energy measurements make explicit.

2.3 Batching, Memory Management, and Inference Serving Systems

At the systems level, Kwon et al. [7] identified key-value cache memory fragmentation as a central bottleneck limiting achievable batch size in LLM serving, and proposed PagedAttention, an operating-system-inspired paging scheme that reduces KV-cache memory waste and thereby enables substantially higher batch sizes and throughput; their evaluation is conducted on datacenter-class GPUs (NVIDIA A100) and focuses on throughput and latency rather than on energy consumption per token, leaving open the question of how KV-cache pressure translates into energy behavior on more memory-constrained, non-datacenter hardware. Pope et al. [13] analyzed the engineering trade-offs of scaling transformer inference across multiple dimensions of hardware partitioning, similarly on TPU-class accelerators, and showed that at long sequence lengths the KV cache, rather than model weight storage, becomes the dominant consumer of device memory; this finding motivates the present study’s context-length factor directly, though Pope et al.’s analysis does not measure energy consumption and does not examine quantized precision as a variable. Argerich and Patiño-Martínez [1] is the most directly comparable prior study: using an EnergyMeter tool built on NVML power sampling, closely analogous to the SystemMonitor component used in the present study, they measured the energy consumption of several open-source LLMs across model architecture, batch size, and quantization level, finding that quantized inference does not uniformly reduce energy consumption because dequantization overhead can offset the memory-related savings. Their study is limited, by the authors’ own account, to short prompts of fewer than ten

tokens and does not treat context length as an experimental factor, which is precisely the limitation the present study is designed to address by adding a third, orthogonal experimental factor — context length ranging from 128 to 1792 tokens — and testing whether the batching and quantization effects reported at short context generalize to substantially longer, more KV-cache-intensive inputs.

2.4 Foundational Model and Evaluation Components

Because the present study evaluates a transformer-based language model, the foundational architecture is attributable to Vaswani et al. [14], and the encoder-based pretraining objective that shaped much of the subsequent large-model landscape, including the decoder-only models evaluated here, is attributable to Devlin et al. [15]. Mixed-precision training, the technique that established FP16 as a standard reduced-precision baseline in deep learning and against which the present study’s INT8 and INT4 comparisons are made, was introduced by Micikevicius et al. [16]. The quality-evaluation protocol used in this study is the Massive Multitask Language Understanding (MMLU) benchmark introduced by Hendrycks et al. [20], which was designed specifically to measure broad factual and reasoning competence across academic subjects using a multiple-choice, letter-scoring format, the same format adopted in the methodology described in Section 3.8. Finally, Zschache and Hartwig [3], in the related but distinct context of text classification inference, demonstrated empirically that energy efficiency and predictive accuracy are separate evaluation dimensions that do not necessarily move together, a finding that directly informs the cautious treatment of the quality results reported in Section 4.6 of the present study.

2.5 Comparison of Prior Studies and Identification of the Research Gap

Table 1 summarizes the scope, method, and principal limitation of the most directly relevant prior studies discussed above, situating the present study’s contribution relative to each.

Table 1. Comparison of the present study with the most closely related prior work. Source: own elaboration.

Study	Factors varied	Energy measured directly?	Context length varied?	Principal limitation relative to the present study
Jacob et al. [4]	Precision (INT8)	No (latency/accuracy only)	No	Vision CNNs, not autoregressive LLMs
Dettmers et al. [8]	Precision (INT8)	No (memory/accuracy only)	No	Does not report energy consumption
Kwon et al. [7]	Batch size, KV-cache management	No (throughput/latency only)	Implicitly, not systematically	Datacenter-class GPU (A100); no energy metric
Pope et al.	Batch size,	No (latency/MFU	Yes	TPU-class hardware;

[13]	sequence length	only)		no quantization factor
Argerich & Patiño-Martínez [1]	Precision, batch size	Yes (direct NVML power sampling)	No (short prompts)	Context length not varied
Niu et al. [21]	Batch size, context length, quantization, model scale	Yes	Yes	Broad benchmark across models and systems rather than a controlled single-model, single-GPU factorial experiment H100/H200
Vellaisamy et al. [22]	Model type, phase, batch size, context length, output length	Yes	Yes	datacenter GPUs and a decomposed energy model rather than the T4 end-to-end setup used here
Present study	Precision, batch size, context length	Yes (direct NVML power sampling)	Yes (128–1792 tokens)	Single model, single GPU generation (see Section 8)

As Table 1 shows, prior work has independently examined several dimensions of LLM inference energy, including quantization, batching, context length, and hardware effects. Argerich and Patiño-Martínez [1] provide the closest methodological comparison because they directly measure GPU energy while varying precision and batch size. More recent benchmark work has expanded the measurement space: Niu et al. [21] introduce TokenPowerBench, a published energy benchmark that varies batch size, context length, quantization, and other deployment factors across multiple model families and hardware settings. Vellaisamy et al. [22] further examine request and token energy as functions of batch size, context length, output length, model type, and inference phase on H100 and H200 GPUs.

The present study therefore does not claim that the general interaction between these factors has never been investigated. Instead, its specific contribution is the controlled factorial examination of precision, batch size, and context length jointly on a single fixed NVIDIA T4 GPU, with repeated end-to-end hardware power measurements and a common model and software configuration across all completed conditions.

2.6 Recent and Concurrent Literature (2025–2026)

Recent published work has increasingly moved toward direct measurement of LLM inference energy rather than relying exclusively on latency, FLOP counts, or hardware power ratings. Niu et al. [21], for example, introduce TokenPowerBench, an energy

benchmark published in the *Proceedings of the AAAI Conference on Artificial Intelligence* in 2026. The benchmark supports systematic variation of batch size, context length, parallelism strategy, and quantization, and evaluates multiple model families ranging from approximately one-billion-parameter models to frontier-scale systems. This broader benchmark perspective reinforces the importance of evaluating energy as an explicit inference metric while also highlighting the methodological distinction between broad benchmarking across heterogeneous systems and the controlled single-model, single-GPU design adopted in the present study.

Vellaisamy et al. [22] provide another particularly relevant comparison. Their study decomposes LLM inference energy into fixed request-level costs and marginal per-output-token costs and evaluates the resulting behavior on NVIDIA H100 and H200 GPUs across model type, inference phase, batch size, context length, and output length. For Llama-3.2-1B on an H200, they report that the energy benefit of batching becomes strongly context-dependent, with the batch-16-to-batch-1 token-energy gain decreasing substantially as context length increases. The work is scheduled for publication at the 2026 IEEE International Symposium on Workload Characterization (IISWC 2026).

This result provides an important external comparison for the batching-by-context interaction observed in the present study. However, the hardware, measurement decomposition, and workload design differ substantially: Vellaisamy et al. evaluate H100/H200 datacenter GPUs and explicitly separate request-level and marginal token-level energy, whereas the present study measures end-to-end energy per output token on a single T4 GPU. The two studies therefore provide complementary rather than interchangeable evidence.

Recent systems research also continues to emphasize the memory consequences of long-context inference. For example, IEEE-published work on long-context inference identifies the KV cache as a major source of memory pressure as context length and batch size increase, particularly in resource-constrained settings. This is consistent with the memory-bound interpretation of the out-of-memory behavior observed in the present study, although such systems work does not by itself establish the energy behavior measured here.

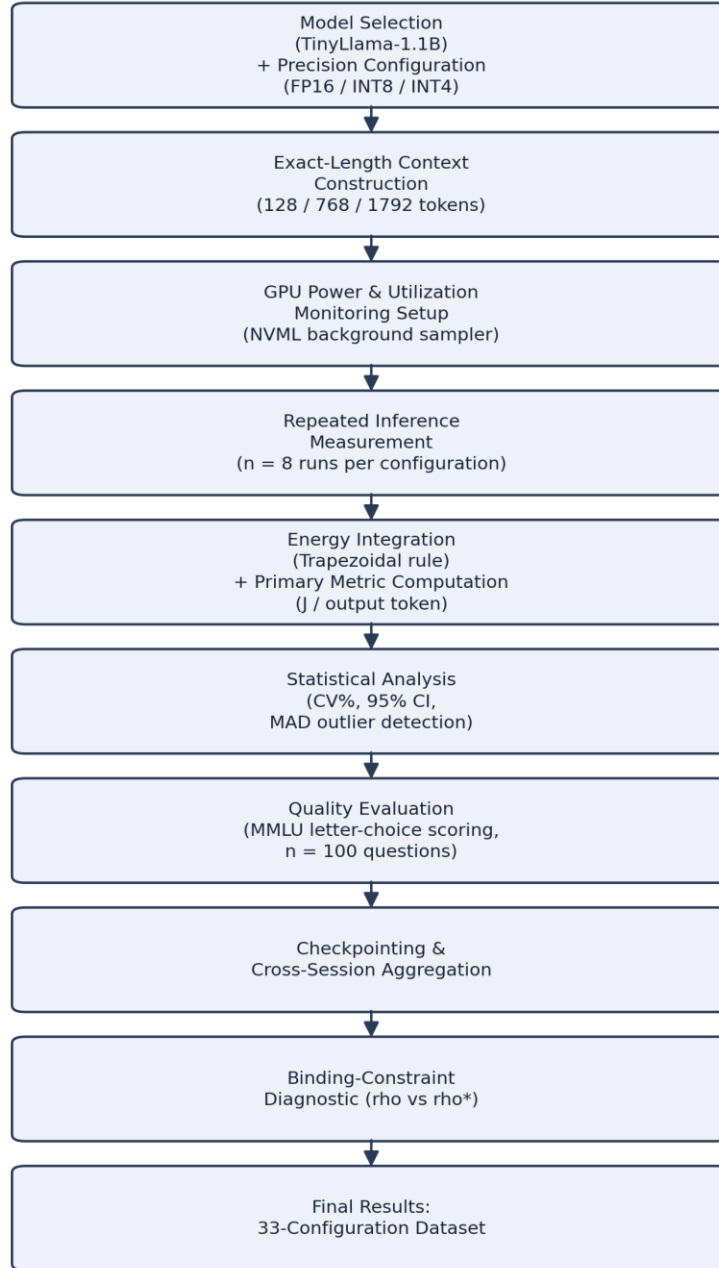
Taken together, the recent literature indicates that LLM inference energy depends on a combination of model characteristics, hardware, context length, batch size, quantization, and serving configuration. The present study therefore makes a deliberately narrower claim: it provides a controlled empirical characterization of the joint effect of three factors—precision, batch size, and context length—on end-to-end energy per output token for TinyLlama-1.1B-Chat running on an NVIDIA T4. Its results should consequently be interpreted as setup-specific evidence rather than as a universal ranking of numerical precisions or inference strategies.

3. Methodology

3.1 Overview of the Measurement Pipeline

The overall measurement pipeline, illustrated in Figure 1, proceeds through model and precision configuration, exact-length context construction, background power and utilization monitoring, repeated inference measurement, energy integration and primary-metric computation, statistical analysis, quality evaluation, cross-session checkpointing, and finally the binding-constraint diagnostic that classifies each configuration as either power-limited or compute-limited. Each stage is described in the subsections that follow.

Figure 1. Block diagram of the experimental measurement pipeline, from model configuration through to the final results dataset. *Source: own elaboration, derived from the experimental implementation described in this study.*



Workflow diagram

3.2 Hardware and Software Environment

All experiments were run on a single NVIDIA T4 GPU (16 GB VRAM), a Turing-architecture accelerator [18], provisioned through Google Colab, using TinyLlama-1.1B-Chat-v1.0 (approximately 1.1 billion parameters) as the model under test, PyTorch together with the HuggingFace Transformers library for model loading and generation, and the bitsandbytes library (version 0.46.1 or later), which implements the eight-bit quantization procedure of

Dettmers et al. [8], for INT8 and INT4 post-training quantization. Power was sampled through the NVIDIA Management Library (NVML). The eager attention implementation was used throughout; FlashAttention 2 was excluded from the experimental matrix for reasons discussed in Section 8.

3.3 Experimental Factors and Design

Three factors were varied in a full factorial design: numerical precision, with three levels corresponding to FP16 (the unquantized baseline, following the reduced-precision convention established by Micikevicius et al. [16]), INT8 (bitsandbytes eight-bit post-training quantization, following the integer-arithmetic quantization scheme formalized by Jacob et al. [4] and implemented for transformers by Dettmers et al. [8]), and INT4 (bitsandbytes four-bit NF4 quantization with FP16 compute dtype); batch size, with four levels of 1, 4, 8, and 16; and context length, with three levels constructed at the exact token level (described in Section 3.5) rather than approximated by repeating a fixed string, and labeled throughout the remainder of this paper as **short** (128 tokens), **medium** (768 tokens), and **long** (1792 tokens). This design yields thirty-six candidate configurations, of which thirty-three completed successfully; the remaining three, all corresponding to batch size 16 combined with the longest context length, failed with a CUDA out-of-memory error regardless of precision level, a result discussed further in Section 4.4 and interpreted in light of the KV-cache memory pressure identified by Kwon et al. [7] and Pope et al. [13] in Section 5.

3.4 Energy Measurement

Power was sampled asynchronously in a background thread at a fixed interval throughout each generation call, using NVML device-level power readings in Watts. Energy was then computed by numerically integrating the resulting power–time series using the trapezoidal rule, so that total energy E is approximated as the sum, over consecutive sample pairs, of the average of the two power readings multiplied by the time elapsed between them. This follows the same underlying principle used by Argerich and Patiño-Martínez [1] in their EnergyMeter tool, which likewise integrates NVML power samples over the duration of a monitored task.

Because cloud-hosted GPUs do not expose facility-level power meters, facility-adjusted energy is reported only as an estimate, obtained by multiplying the measured device energy by an assumed Power Usage Effectiveness (PUE) ratio, a metric whose typical range across operational data centers is documented in the annual Uptime Institute Global Data Center Survey [19]. Rather than committing to a single, unverifiable PUE value, this study instead sweeps PUE across the range 1.1 to 1.6 and verifies, in Section 4.5, that the relative ranking of configurations by energy efficiency does not change across this range.

3.5 Exact-Length Context Construction

To ensure that the three context-length conditions correspond to precisely the intended number of input tokens rather than an approximation, each input sequence was constructed by tokenizing a fixed filler sentence, tiling it to the required length at the token level, and appending a fixed instruction segment, so that the total token count matched the

target length exactly and every context-length condition still posed the same underlying question to the model.

3.6 Rejection of Differential Decode-Energy Estimation

An earlier iteration of this measurement pipeline attempted to isolate decode-phase energy from prefill-phase energy by subtracting the energy measured for generating a single token from the energy measured for generating the full target number of tokens, using two separate generation calls per measurement. This approach was validated directly by measuring total energy at several intermediate generation lengths and testing whether energy grew linearly with the number of generated tokens, as it must if the subtraction reliably isolates a stable per-token decode cost. The resulting linear fit produced an R-squared value of 0.737, below the threshold of 0.90 adopted for this study as the minimum acceptable evidence of linearity, and a regression slope that was negative, implying that generating more tokens corresponded to lower measured energy, a result that is physically implausible and indicates that the noise introduced by combining two independently sampled short-duration power-integration windows exceeded the signal the subtraction was intended to isolate. As a direct consequence of this diagnostic, all results reported in this study use end-to-end energy per output token, computed from a single generation call covering both the prefill and decode phases together, as the primary metric.

3.7 Primary Metric and Statistical Treatment

The primary metric reported throughout this study is energy per output token, defined as total measured device energy divided by the number of output tokens generated in that call. For each of the thirty-three completed configurations, eight repeated measurements were collected after two discarded warm-up runs, and the median and mean of the resulting energy-per-token values are reported together with the coefficient of variation, expressed as the ratio of the standard deviation to the mean. A ninety-five percent confidence interval was computed both through the standard Student's t-distribution and through a two-thousand-resample bootstrap of the median. Individual runs were flagged as outliers using the median absolute deviation criterion, whereby a run was excluded from summary statistics if its deviation from the configuration's median exceeded three times the median absolute deviation, though all raw measurements, including flagged outliers, were retained in the underlying dataset.

3.8 Quality Evaluation

Output quality was evaluated using the standard letter-choice scoring protocol associated with the Massive Multitask Language Understanding (MMLU) benchmark of Hendrycks et al. [20]: each question was formatted with lettered answer choices followed by the word "Answer:", a single forward pass was run, and the model's predicted answer was taken as the choice among the four letters whose next-token logit was highest. One hundred fixed questions, drawn with a fixed random seed from the MMLU dataset, were used identically across all configurations. Because output quality depends only on model weights, precision, and attention implementation, and not on batch size or on the context-length manipulation used for the energy measurements, quality was evaluated once per precision-and-attention

combination rather than once for every full configuration. It is noted here, and returned to in Section 4.6, that a four-choice multiple-choice format has a random-guessing floor of 25 percent accuracy, and the accuracies obtained in this study (Section 4.6) sit close to that floor; this is an expected consequence of evaluating a 1.1-billion-parameter model, which is substantially smaller than the models for which MMLU is typically reported, and it materially limits how much this particular probe can say about quality differences between precision levels, independent of the sample-size concern addressed statistically in Section 4.6.

3.9 Binding-Constraint Diagnostic

Following the conceptual distinction between compute-bound and power-bound inference regimes, each configuration’s measured average power draw and measured energy per token were compared against three tiers of assumed hardware compute capability: the manufacturer-published theoretical peak floating-point throughput for the T4 GPU, drawn from the NVIDIA Turing architecture specification [18]; a directly measured throughput obtained by benchmarking a large dense matrix multiplication on the same device, following the general efficiency-characterization approach surveyed by Sze et al. [10]; and a model-based proxy obtained by multiplying twice the model’s parameter count by the measured tokens-per-second throughput of the configuration in question. A configuration was classified as power-limited when the ratio of average power to assumed hardware capability was smaller than the ratio of measured energy per token to compute per token, and as compute-limited otherwise. The third tier is explicitly a model-based approximation rather than a direct hardware measurement, since it does not account for the fact that the computational cost of self-attention grows with context length while the dense-model approximation used here treats compute per token as constant.

3.10 Reproducibility and Checkpointing

Because cloud GPU sessions of the kind used in this study are subject to interruption, each configuration’s summary result was written to a checkpoint file immediately upon completion, and the measurement procedure was designed to detect and skip any configuration already present in that file when re-invoked.

4. Results

4.1 Primary Metric: Energy per Output Token

Table 2 reports the median energy per output token, in joules, for all thirty-three completed configurations, together with the corresponding quality score for each precision level. *Source: own elaboration, based on the experimental measurements collected for this study.*

Table 2. Median energy per output token (J/token) by precision, batch size, and context length.

Precision	Batch	Short (128 tok)	Medium (768 tok)	Long (1792 tok)	Quality (MMLU)
FP16	1	3.71	9.93	36.99	22%
FP16	4	2.20	9.46	38.10	22%
FP16	8	1.25	9.98	43.02	22%
FP16	16	1.30	10.40	OOM	22%
INT8	1	7.36	20.06	64.27	21%
INT8	4	3.02	17.22	59.82	21%
INT8	8	2.84	15.94	62.21	21%
INT8	16	2.20	16.14	OOM	21%
INT4	1	6.39	17.74	63.80	29%
INT4	4	2.50	16.53	60.40	29%
INT4	8	1.86	15.77	61.79	29%
INT4	16	1.72	15.99	OOM	29%

Coefficient of variation across the thirty-three configurations was generally low, with a median value of approximately three to five percent; Table 3 reports the coefficient of variation for every individual configuration so that the point estimates in Table 2 can be judged against their measurement dispersion rather than read as exact values.

Table 3. Coefficient of variation (CV%) of energy per output token, by configuration (n = 8 repeated measurements each). *Source: own elaboration, based on the experimental measurements collected for this study.*

Precision	Batch	Short (128 tok)	Medium (768 tok)	Long (1792 tok)
FP16	1	19.20%	11.62%	5.16%
FP16	4	15.33%	0.65%	11.77%
FP16	8	7.95%	1.26%	1.78%
FP16	16	2.11%	0.42%	OOM
INT8	1	4.00%	2.67%	2.05%
INT8	4	12.81%	0.98%	1.30%
INT8	8	6.30%	0.86%	0.83%
INT8	16	5.49%	0.39%	OOM
INT4	1	4.07%	4.79%	1.94%
INT4	4	7.02%	0.63%	0.88%
INT4	8	11.18%	0.40%	0.83%
INT4	16	6.68%	0.59%	OOM

The highest dispersion in the entire dataset, 19.20 percent, occurs specifically at the FP16, batch-1, short-context configuration, and dispersion is visibly higher across the board at

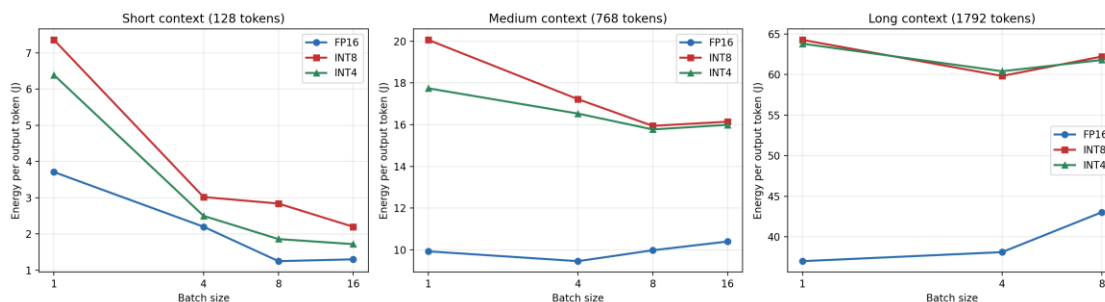
batch size 1 than at larger batch sizes; this is consistent with the shorter absolute duration of low-batch, short-context runs yielding fewer power samples per measurement window (Section 3.4), and readers comparing point estimates at batch size 1 in Table 2 should weight them accordingly against the corresponding row in Table 3.

4.2 Quantization Did Not Reduce Energy Consumption

In every batch-by-context cell reported in Table 2, FP16 achieved energy per token equal to or lower than at least one of the two quantized variants, and in the majority of cells FP16 was the single most efficient configuration overall. INT8 was the least efficient precision level in nearly every cell despite occupying the intermediate position between FP16 and INT4. At batch size 1 with short context, FP16 required 3.71 joules per token, compared with 6.39 for INT4 (an increase of approximately 72 percent) and 7.36 for INT8 (an increase of approximately 98 percent); the coefficient of variation at this specific cell is 19.20 percent for FP16 and 4.07 percent for INT4 (Table 3), and even allowing generously for this dispersion, the two distributions do not visibly overlap, so the direction of this particular comparison is not an artifact of measurement noise. This pattern is directly consistent with the mechanism described by Argerich and Patiño-Martínez [1], who report that inference latency, and consequently energy consumption, can increase under quantization due to dequantization overhead, an effect not guaranteed to be offset by the reduced memory footprint reported by Dettmers et al. [8] and Jacob et al. [4].

4.3 The Batching Benefit Is Context-Dependent, and Reverses at Long Context

Figure 2 plots energy per output token against batch size, separately for short, medium, and long context, for all three precision levels. Source: own elaboration, based on the experimental measurements reported in Table 2.



Energy vs batch size interaction

At short context (128 tokens), increasing batch size from 1 to 8 or 16 reduced FP16 energy per token by approximately 65 percent, from 3.71 to between 1.25 and 1.30 joules per token, and comparable proportional reductions are visible for INT8 and INT4 in the left panel of Figure 2. At medium context (768 tokens), the center panel of Figure 2 shows a materially different pattern: energy per token remains within a narrow band across all four batch sizes tested for every precision level, with no configuration showing a reduction larger than approximately five percent. At long context (1792 tokens), the rightmost panel of Figure 2 reveals a stronger context-dependent effect: for FP16, energy per token rises as

batch size increases, from 36.99 joules per token at batch size 1, to 38.10 at batch size 4, to 43.02 at batch size 8—an increase of approximately 16 percent over the batch-1 baseline.

INT8 and INT4 show a milder version of the same pattern, improving from batch size 1 to 4 before partially eroding at batch size 8. These observations indicate that the energy benefit of batching is not invariant across context lengths. Rather, the direction and magnitude of the effect depend on the interaction between batch size and the amount of contextual state maintained during inference.

This interpretation is consistent with recent independent measurements by Vellaisamy et al. [22], who report that the energy benefit of batching becomes increasingly limited as context length increases on H100 and H200 GPUs. Their study uses a different hardware generation and an energy decomposition different from the end-to-end measurement used here, but the direction of the context-dependent batching effect provides an external comparison for the pattern observed in Figure 2.

The present result should therefore be interpreted as evidence that batching efficiency is workload-dependent rather than as evidence that batching universally becomes counterproductive at long context. In particular, the FP16 reversal observed here is specific to the tested model, hardware, software configuration, context range, and output-generation workload

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4.4 Memory-Bound Failure Is Precision-Independent

All three out-of-memory failures occurred at the identical cell, batch size 16 combined with the longest context length, across all three precision levels, despite INT4 nominally requiring substantially less memory for model weights than FP16. This is consistent with the argument advanced by Kwon et al. [7] and Pope et al. [13] that, at large batch size and long sequence length, the key-value cache rather than the model weights becomes the dominant consumer of GPU memory, a factor that weight-precision reduction alone does not address.

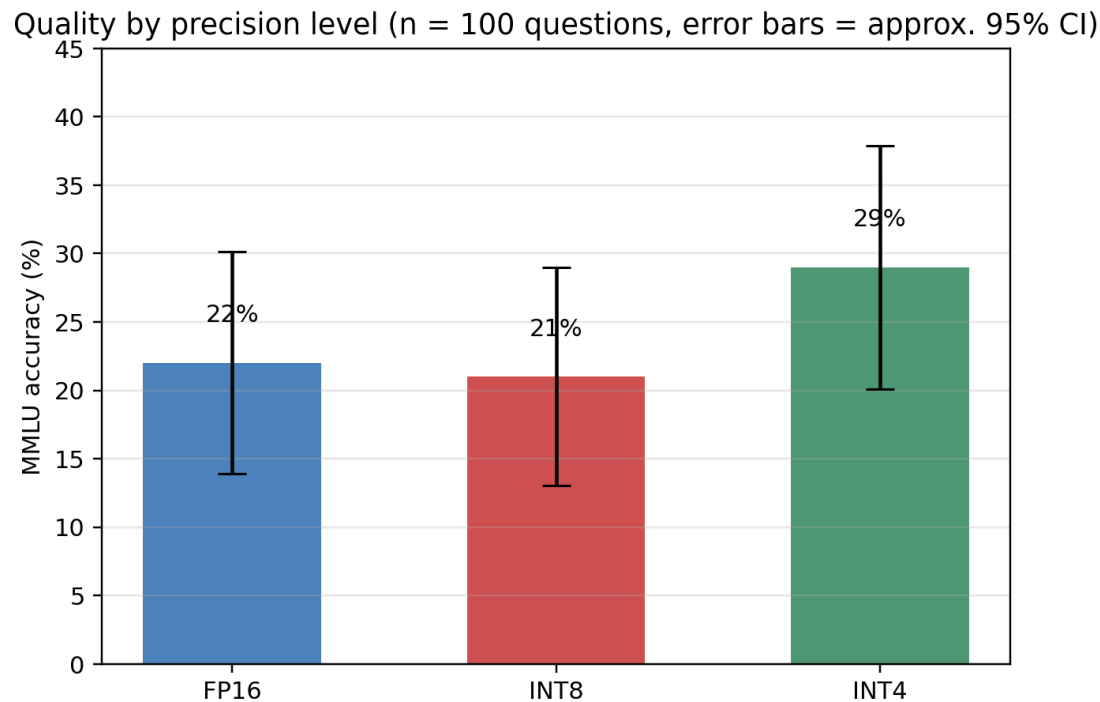
4.5 PUE Robustness

Configuration rankings by facility-adjusted energy, computed at PUE values of 1.1, 1.2, 1.4, and 1.6 — a range consistent with the values reported in recent Uptime Institute industry surveys [19] — were found to be stable across this entire range.

4.6 Quality

Figure 3 presents the MMLU accuracy achieved by each precision level, with error bars indicating an approximate 95 percent confidence interval for a binomial proportion at the sample size used in this study. *Source: own elaboration, based on the experimental measurements described in Section 3.8.*

Figure 3. MMLU accuracy by precision level, with approximate 95 percent confidence intervals (n = 100 questions).



Quality by precision

FP16, INT8, and INT4 achieved MMLU accuracies of 22, 21, and 29 percent respectively, values that were stable across repeated pipeline executions. As Figure 3 makes visually explicit, the confidence interval around each of the three estimates is wide relative to the differences between them: the interval for INT4, spanning roughly 20 to 38 percent, overlaps substantially with the interval for FP16, spanning roughly 14 to 30 percent, and both overlap with the interval for INT8. This overlap is the statistical reason the seven-percentage-point gap between FP16 and INT4 cannot, on the evidence of this dataset alone, be attributed with confidence to a genuine quality difference induced by quantization rather than to sampling variability in which specific questions happened to be answered correctly at each precision level. Consistent with the finding of Zschache and Hartwig [3] that energy efficiency and predictive accuracy are separate evaluation dimensions that do not necessarily move together, no claim is made here that INT4’s higher observed point estimate reflects a general quality advantage.

5. Discussion

5.1 Interpretation of the Quantization Result

The finding that FP16 outperformed both quantized precisions in nearly every configuration runs counter to the intuitive expectation, common in practitioner discussions

and in some of the model-compression literature reviewed in Section 2.2, that fewer bits per weight should translate directly into lower energy consumption. Unpacking why this expectation fails requires separating two effects that quantization changes simultaneously and that pull in opposite directions. On one hand, storing each weight in fewer bits reduces the volume of data that must be moved from GPU memory to the compute units for every matrix multiplication, and since memory movement itself consumes energy, this effect should, in isolation, reduce energy per token as the bit-width decreases. On the other hand, the arithmetic units on the T4 GPU compute natively in floating-point and integer formats at specific fixed widths; a value stored at INT8 or INT4 precision cannot be multiplied directly in that reduced format on this architecture and must first be *dequantized*, that is, converted back to a higher-precision floating-point representation, before the multiplication is performed. This dequantization step is itself a computation that consumes time and energy, and it is repeated for every weight, in every layer, at every one of the many sequential decoding steps that autoregressive generation requires. The net energy effect of quantization is therefore the difference between the energy saved by moving less data and the energy spent on repeated dequantization, and the sign of this difference depends on the balance between the two, which in turn depends on the specific hardware’s support for native low-precision arithmetic.

This reasoning, combined with the direct empirical parallel reported by Argerich and Patiño-Martínez [1] on a different model family using the same class of NVML-based measurement, leads to the interpretation that on the T4 GPU used in this study, the dequantization cost outweighs the data-movement saving, so that the net effect of quantization is an increase rather than a decrease in energy per token. This stands in contrast to the results typically reported for quantization on newer, datacenter-class GPUs that include dedicated low-precision tensor-core paths capable of operating on INT8 or INT4 data natively, without a separate dequantization step; in that hardware context, the balance of the same two effects described above would be expected to tilt the other way, toward a net energy saving, which is consistent with why the wider literature surveyed in Section 2.2, largely benchmarked on such hardware, generally reports quantization as beneficial. The present result should therefore be read not as a contradiction of that literature, but as evidence that the reported benefit is conditional on hardware support for low-precision arithmetic that is present on flagship accelerators and not present, or not fully present, on the T4.

A second observation requiring explanation is why INT8, rather than INT4, was the least energy-efficient precision level in nearly every configuration in Table 2, which is at first counter-intuitive since INT8 retains more numerical precision and might be assumed to require a “smaller” or cheaper dequantization step than INT4. The explanation most consistent with the bitsandbytes implementation used in this study is that the INT8 and INT4 (NF4) code paths in that library are not simply scaled versions of the same procedure: the INT4 NF4 scheme is specifically optimized as a fused, single-pass dequantization kernel designed for the common case of decoding one or a few tokens at a time, whereas the INT8 implementation used here follows a more general vector-wise scheme that was not originally optimized for this specific decoding pattern. This illustrates a broader methodological point that follows directly from this result: energy cost cannot be inferred

from bit-width alone, because bit-width interacts with the specific kernel implementation used to execute that precision on a given piece of hardware, and two different quantization schemes nominally operating at different precisions can therefore be ranked in either order depending on implementation details that are invisible from the precision label alone.

5.2 Interpretation of the Batching-by-Context Interaction

The progression from a strong batching benefit at short context, to a negligible benefit at medium context, to an increase in energy per token for FP16 at long context is one of the most important findings of this study. It should, however, be interpreted as a setup-specific interaction rather than as a universal law of batching.

A useful way to interpret the observation is to distinguish between costs that can be amortized across multiple requests and costs that increase with the amount of contextual state processed during decoding. At short context and low batch size, fixed work associated with initiating inference and processing model weights represents a relatively large portion of the energy consumed per output token. Increasing the batch size allows more sequences to share the same model execution and can therefore improve hardware utilization and reduce the energy attributed to each output token.

As context length grows, the KV cache becomes increasingly important. Each sequence maintains its own cached keys and values, and the amount of cached information grows with the number of previous tokens. Increasing batch size therefore increases the total amount of KV-cache state that must be maintained and accessed. The resulting memory traffic can reduce or offset the utilization benefit obtained from batching.

This interpretation is consistent with established systems research on LLM serving. Kwon et al. [7] identify KV-cache memory as a major constraint on achievable batch size, while Pope et al. [13] show that KV-cache requirements become increasingly important as sequence length grows. The more recent energy measurements of Vellaisamy et al. [22] provide an independent empirical indication that batching gains in energy per token become context-bounded on different GPU hardware.

The present dataset provides an additional observation within this framework: at the longest tested context, FP16 does not merely lose its batching advantage but shows an increase in energy per output token as batch size rises from 1 to 8. This sign reversal is an empirical property of the tested configuration and should not be generalized to all models or GPUs.

The milder pattern observed for INT8 and INT4 is not fully explained by the present dataset. One possible explanation is that the reduced model-weight footprint of the quantized configurations changes the balance between weight movement and KV-cache traffic. However, this explanation is not directly measured in the present experiment and is therefore treated as a hypothesis rather than a demonstrated mechanism. Testing it would

require direct measurements of memory bandwidth, cache traffic, and attention-phase energy across a wider range of context lengths.

5.3 Interpretation of the Quality Result

The observed MMLU accuracies of 22, 21, and 29 percent for FP16, INT8, and INT4 respectively, shown together with their overlapping confidence intervals in Figure 3, should not be read, on their own, as evidence that INT4 quantization improved model quality relative to the unquantized baseline. Two distinct explanations are consistent with the pattern visible in Figure 3, and the present dataset does not allow them to be distinguished. The first is that the true underlying accuracy of the three precision levels is in fact very similar, and the observed spread of seven percentage points is simply the ordinary sampling noise that arises whenever a capability is measured on a fixed set of only one hundred questions rather than on the full MMLU test set; under this explanation, a larger question sample would be expected to pull all three estimates closer together. The second is that a small genuine quality difference does exist, in either direction, but is too small relative to the measurement’s statistical power to be distinguished from noise at this sample size. Distinguishing between these two explanations would require either a substantially larger question sample or a paired statistical test, such as McNemar’s test, applied to the subset of questions on which the precision levels disagree; neither was performed as part of this study, and the quality result is accordingly reported descriptively rather than as a validated finding. This caution is reinforced by Zschache and Hartwig [3], who demonstrate in a different but methodologically related context that energy efficiency and predictive accuracy are separate axes of evaluation that can move independently and even in counter-intuitive directions, meaning that a favorable energy result for one precision level, such as the batching-related gains discussed in Section 5.2, carries no implication whatsoever about that precision level’s expected accuracy, and the two properties must always be evaluated and reported separately, as they are in Table 2 and Figure 3 of this study.

5.4 Practical and Theoretical Implications

For practitioners deploying LLMs on GPUs outside the datacenter tier, the practical implication is that quantization should not be assumed to reduce energy consumption by default. Its effect should instead be measured for the specific combination of model, hardware, software stack, and workload. The present measurements show that quantization can increase energy per output token under the tested T4 configuration, even though broader benchmarking studies demonstrate that quantization can provide energy benefits under other hardware and workload conditions. TokenPowerBench, for example, explicitly evaluates quantization alongside batch size and context length across heterogeneous inference configurations, reinforcing the importance of evaluating these factors jointly rather than in isolation.

The batching results similarly indicate that energy-efficiency claims should be scoped to the context-length regime in which they were measured. The strong reduction observed at 128 tokens, the near-plateau at 768 tokens, and the FP16 increase at 1792 tokens demonstrate that the same optimization can have different energy consequences as workload characteristics change.

These findings do not imply that batching is generally inefficient at long context. Rather, they show that the energy benefit of batching is conditional. This interpretation is consistent with recent independent measurements on H100 and H200 GPUs, where batching gains in token-normalized energy also diminish as context length increases.

The broader theoretical implication is that LLM inference energy should be treated as a systems-level property emerging from the interaction between model architecture, numerical precision, memory behavior, batch size, context length, and hardware implementation. No single factor can therefore be assumed to determine energy efficiency independently of the others.

6. Conclusion

This study measured, through a controlled, checkpointed, and repeated measurement pipeline, the energy per output token of LLM inference across thirty-three completed configurations spanning three precision levels, four batch sizes, and three context lengths on a single NVIDIA T4 GPU.

The main contributions of this study are as follows:

- Direct empirical evidence, obtained through hardware power measurement rather than proxy metrics, showing that unquantized FP16 inference exhibited lower energy per output token than INT8 and INT4 in most of the tested configurations on the evaluated T4 setup. This result demonstrates that lower numerical precision does not automatically imply lower energy per output token.
- Demonstration of a strong interaction between batch size and context length. A substantial batching-related energy benefit was observed at short context (128 tokens), the benefit weakened substantially at medium context (768 tokens), and FP16 showed an increase in energy per output token as batch size increased at long context (1792 tokens). This observation indicates that the energy benefit of batching can depend strongly on context length and system configuration.
- Identification of a precision-independent memory-bound failure threshold at batch size 16 combined with the longest tested context length. The fact that all three precision levels failed at the same batch-context combination provides evidence that KV-cache growth, rather than model-weight storage alone, was an important memory constraint under the tested conditions.
- A documented methodological finding concerning energy measurement: differential estimation of decode-phase energy through subtraction of separately measured short-duration generation windows produced unreliable results at the workload

scale used here ($R^2 = 0.737$ and a physically implausible negative regression slope). This motivated the adoption of end-to-end energy per output token as the primary metric.

These findings should be interpreted as setup-specific empirical evidence. The study does not establish that FP16 is universally more energy-efficient than quantized inference, nor that batching universally becomes counterproductive at long context. Instead, it demonstrates that the energy consequences of these optimizations depend on the interaction between numerical precision, batch size, context length, hardware, and implementation.

7. Recommendations and Future Work

For practitioners selecting an inference configuration on non-datacenter-class GPUs, the immediate recommendation arising from this study is to measure energy per output token directly for the specific model, precision, batch size, and expected context-length range of the target deployment rather than assuming that a lower-precision configuration is necessarily more energy-efficient.

For the research community, several directions for future work follow directly from the limitations discussed in Section 8.

The first is to repeat the experimental design on additional model families and GPU generations. In particular, comparing the NVIDIA T4 with a more recent GPU possessing stronger native low-precision support would help determine whether the energy behavior observed here is specific to the T4 generation and the particular quantization implementation used.

The second is to extend the context-length range beyond 1792 tokens using a memory-efficient serving approach such as PagedAttention [7]. This would allow the study to determine whether the different batching patterns observed for FP16, INT8, and INT4 persist, converge, or change at longer contexts.

The third is to develop a validated method for separating prefill-phase and decode-phase energy within a single generation call. Such a measurement would make it possible to determine more precisely which phase contributes most strongly to the context-length interaction observed in this study.

The fourth is to supplement the near-floor MMLU accuracy probe with an additional quality metric such as perplexity on a held-out text corpus. This would provide a wider dynamic range for evaluating quality differences between precision levels on relatively small language models.

Finally, future studies should combine direct energy measurement with GPU memory utilization, memory bandwidth, compute utilization, and kernel-level profiling. Such measurements would allow the mechanisms underlying the observed precision and batching effects to be tested directly rather than inferred from aggregate energy measurements.

8. Limitations

This study evaluates a single model, TinyLlama-1.1B, on a single GPU architecture, the NVIDIA T4. Consequently, generalization of the findings to other model families, model sizes, GPU generations, and software implementations is not established by the present data.

The study also uses a specific quantization implementation through bitsandbytes. Therefore, the observed INT8 and INT4 energy behavior should not be interpreted as representative of every possible INT8 or INT4 implementation. Quantization performance depends not only on numerical precision but also on kernel implementation, memory access patterns, hardware support, and the surrounding inference software stack.

FlashAttention 2 was excluded from the experimental matrix due to installation reliability constraints in the execution environment, so all reported results use the eager attention implementation. This limits the extent to which the measured context-length behavior can be generalized to alternative attention implementations.

The number of tokens generated per measurement differed between the majority of configurations, at one hundred tokens, and the configurations combining batch size eight or larger with the longest context length, at thirty tokens, the latter having been reduced as a precautionary measure against out-of-memory failure. Although the primary metric is normalized per output token, this asymmetry is disclosed explicitly rather than treated as fully controlled for.

The PUE value used for facility-adjusted energy estimates is an assumption informed by industry survey data [19] rather than a direct facility-level measurement. This limitation is mitigated through the sensitivity analysis reported in Section 4.5.

The third tier of the binding-constraint diagnostic is a model-based approximation of compute per token rather than a directly measured FLOP count, and does not account for the increasing computational contribution of self-attention as context length grows.

The quality-evaluation probe evaluates a 1.1-billion-parameter model on MMLU, whose four-choice format has a random-guessing floor of 25 percent. The observed accuracies therefore provide limited evidence for distinguishing genuine precision-induced quality differences. A larger evaluation set and complementary metrics would provide stronger evidence.

Finally, eight repeated measurements per configuration and one hundred quality-evaluation questions provide useful estimates of measurement variability but do not constitute a large-sample guarantee. The findings should therefore be interpreted as empirical evidence for the tested experimental conditions rather than as universal performance laws.

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